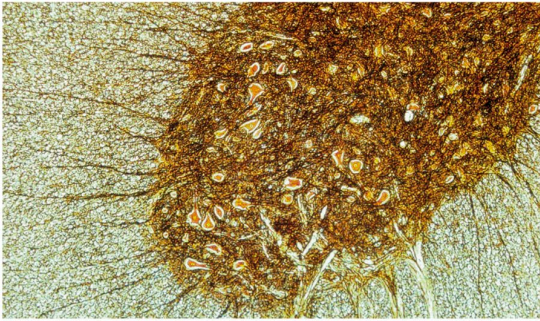




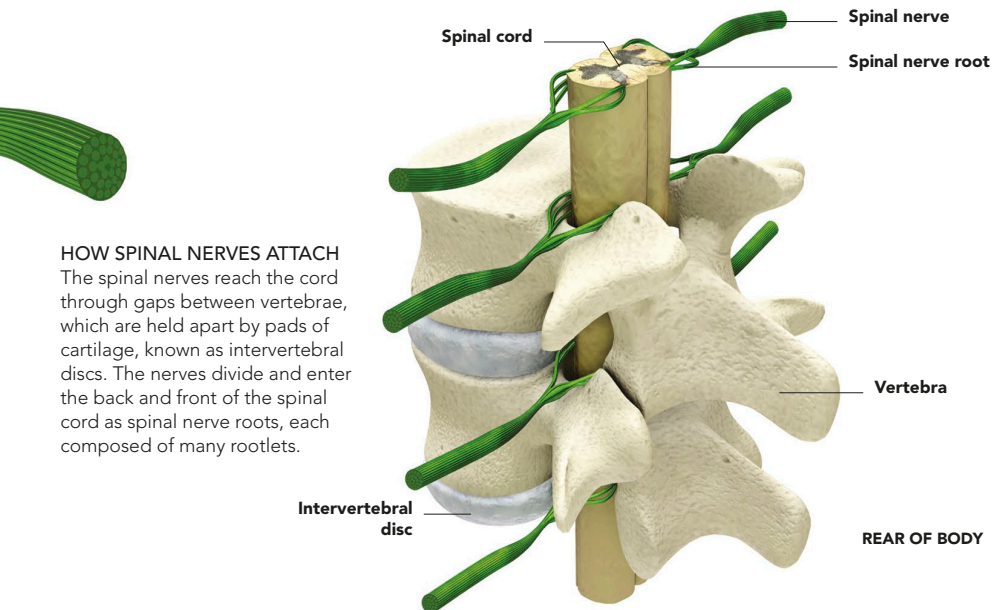
SPINAL GREY MATTER

This microscopic view of a cross-section through the spinal cord shows a brown-stained "wing" of the butterfly-shaped grey matter, which lies at the cord's centre.

NERVE CROSSOVER

Bundles of nerve fibres (axons) in the left and right sides of the spinal cord do not all pass straight up into the left and right sides of the brain. In the uppermost portion of the spinal cord and the lower brainstem (the part called the medulla; see p.93), many of the fibres cross over, or decussate, to the other side – left to right, and right to left. This means that nerve signals about, for example, touch sensations on the left side of the body reach the

touch centre (somatosensory cortex) on the right side of the brain. Likewise, motor signals from the right motor cortex in the brain and the right side of the cerebellum travel to the muscles on the left side of the body. Different major bundles, or tracts, of fibres decussate at slightly different levels. About one-tenth of those that cross over do so in the upper spinal cord, and the remainder cross over in the medulla.



HOW SPINAL NERVES ATTACH

The spinal nerves reach the cord through gaps between vertebrae, which are held apart by pads of cartilage, known as intervertebral discs. The nerves divide and enter the back and front of the spinal cord as spinal nerve roots, each composed of many rootlets.